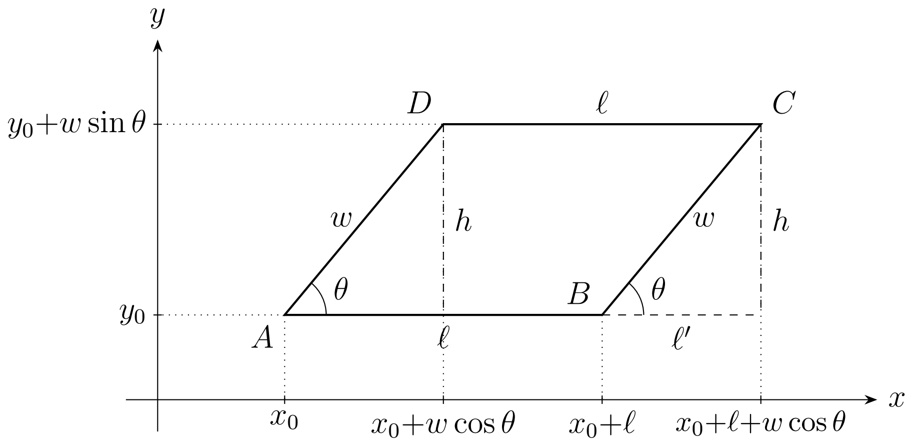

Relations between ℓ', h and w:

$$\cos\theta = \frac{\ell'}{w} \quad \Rightarrow \quad \ell' = w\cos\theta$$

$$\sin\theta = \frac{h}{w} \quad \Rightarrow \quad h = w\sin\theta$$

Vertex coordinates:

$$\begin{aligned}
A &= (x_0,\ y_0)\\
B &= (x_0+\ell,\ y_0)\\
C &= (x_0+\ell+w\cos\theta,\ y_0+w\sin\theta)\\
D &= (x_0+w\cos\theta,\ y_0+w\sin\theta)
\end{aligned}$$